

Jump-Diffusion Models in Financial Markets

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Questions of Interest

1. What are Jump-Diffusion models and how do they work?
2. How do they compare to the Black-Scholes model?

History

- ▶ In a 1976 paper, **Robert C. Merton** first explored extending the Black-Scholes model with a jump term.
 - The same Merton of the sometimes otherwise named Black-Scholes-Merton model.
- ▶ What was he trying to address?
 - At the time, the **heavier tails/higher peak** of financial asset returns than presented by the Normal distribution.
 - Later, jump terms were also determined to help address the **Volatility smile**.

S&P 500 Returns

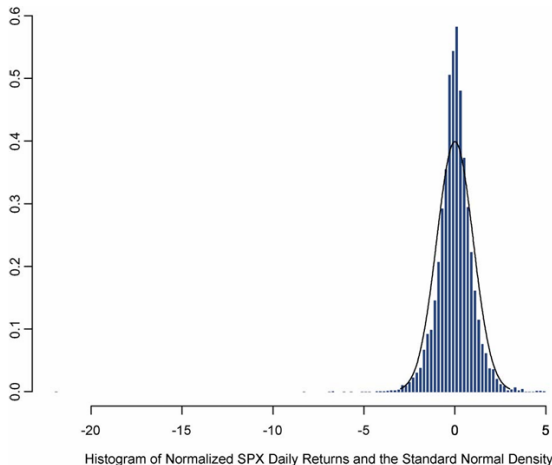


Fig. 2. Comparison of the histogram of the normalized daily returns of S&P 500 index (from Jan 2, 1980 to Dec 31, 2005) and the density of $N(0, 1)$. The feature of a high peak and two heavy tails (i.e. the leptokurtic feature) is quite evident.

[Kou07]

Problem #1: Leptokurtic Distribution

- ▶ Above, we see the S&P 500 returns reflecting heavier tails and a higher peak than the overlaid normal distribution [Kou07].
 - It turns out that this is observed for most financial asset prices.
 - Named a **Leptokurtic distribution**, having large kurtosis.
- ▶ However, the Black-Scholes model assumes Normally distributed increments, which translates to the returns via linearity.

Problem #2: Volatility Smile

- ▶ Assuming we know the parameters in a put/call option pricing formula (e.g. Black-Scholes), we can define an inverse function mapping from a given option price to the volatility parameter.
 - This is the **implied volatility**, $\sigma(T, K)$.
- ▶ Implied volatilities can be used to price options with different strike prices and maturity times.
 - Useful when historical volatilities do not reflect current conditions.
- ▶ In practice, plotting implied volatilities against strike prices produces a curve resembling a "**volatility smile**".
 - However, the Black-Scholes model utilizes Geometric Brownian Motion which assumes the implied volatilities are the same for all options with the same underlying asset.

Poisson Process

- ▶ How do Jump-Diffusion models address these problems?
 - **Jumps** - random, abrupt changes in the underlying asset price.
- ▶ **Poisson Process**: a stochastic process, counting the number of random times, modelled by exponential random variables, occurring within a period.

$$N_t = \sum_{n \geq 1} 1_{t \geq T_n},$$

where $T_n = \sum_{i=1}^n \tau_i$ and $(\tau_i)_{i \geq 1}$ are i.i.d. $Exponential(\lambda)$. λ is called the **intensity** of N_t .

- ▶ **Compound Poisson Process**: using N_t as the number of jumps and another independent sequence of i.i.d. RVs $(Y_i)_{i \geq 1}$ to model jump sizes, we get:

$$X_t = \sum_{i=1}^{N_t} Y_i$$

Jump-Diffusion Models

Adding a Compound Poisson Process to Geometric Brownian Motion gives us:

Jump-Diffusion Models

$$\frac{dS_t}{S_t} = \mu dt + \sigma dW_t + \sum_{i=1}^{dN_t} (Y_i - 1)$$

$$\rightarrow S_t = S_0 \exp\left[\left(\mu - \frac{\sigma^2}{2}\right)t + \sigma W_t\right] \prod_{i=1}^{N_t} Y_i, \text{ by It\^o's Lemma.}$$

where:

- ▶ W_t is standard Brownian motion,
- ▶ $(N_t)_{t \geq 0}$ is the Poisson process counting the number of jumps with intensity λ , and
- ▶ $(Y_i)_{i \geq 1}$ are i.i.d. random variables modelling jump sizes.

Merton vs. Kou Models

Merton

- ▶ The log of **jump sizes**, $(\log(Y_i))_{i \geq 1}$, are i.i.d. $Normal(m, \delta^2)$.
 - $(Y_i)_{i \geq 1}$ are i.i.d. *Lognormal*.
- ▶ **Parameters:** σ - diffusion (volatility), λ - jump intensity (frequency), μ - mean jump size, and δ - jump size standard deviation.

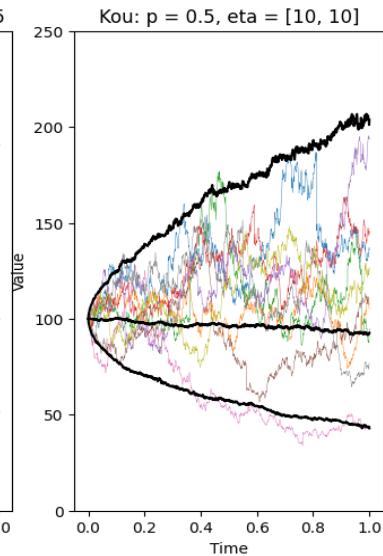
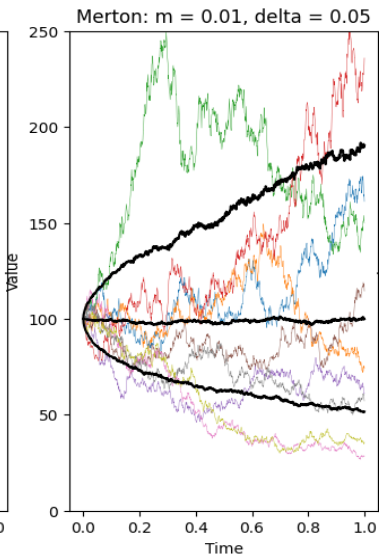
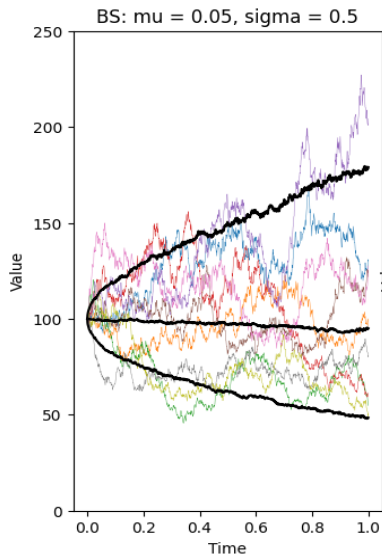
Kou

- ▶ The log of **jump sizes**, $(\log(Y_i))_{i \geq 1}$, are i.i.d. from an asymmetric double exponential distribution.

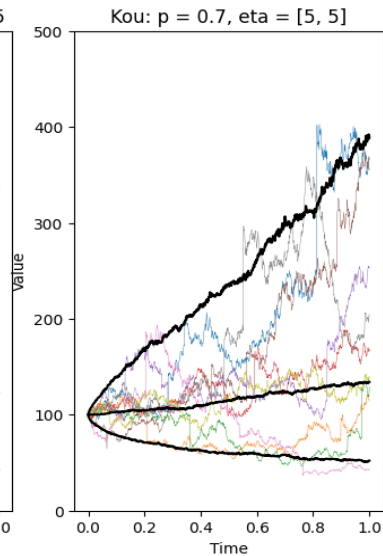
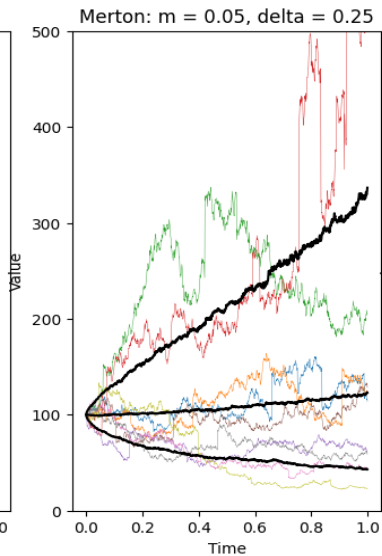
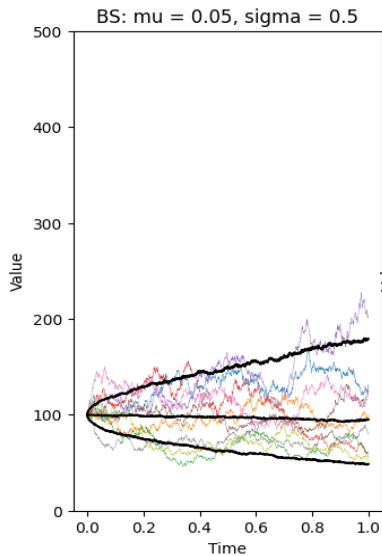
$$f(x) = p\eta_1 e^{-\eta_1 x} 1_{x \geq 0} + (1-p)\eta_2 e^{-\eta_2 x} 1_{x < 0}$$

- ▶ **Parameters:** σ - diffusion (volatility), λ - jump intensity (frequency), p - probability of positive (vs. negative) jump, and $\frac{1}{\eta_1}, \frac{1}{\eta_2} < 1$ - mean positive/negative jump sizes.

Sample Paths #1 - Moderate Parameters



Sample Paths #2 - Large Parameters



Intuition

As suggested by **Steven Kou** [Kou07]:

- ▶ It's been observed that markets tend to have both over- and under-reactions to good and bad news.
 - "One may interpret the jump part of the model as the market response to outside news."
 - i.e. Without any external information, the asset price follows a geometric Brownian motion. External market information arrives according to a Poisson process.
- ▶ "... the double exponential jump-diffusion model can be interpreted as an attempt to build a simple model, within the traditional random walk and efficient market framework, to incorporate investors' sentiment."
 - This same intuition can be applied to the Merton model.

Option Pricing

Jump-Diffusion Model

$$dB_t = rB_t dt, B_0 = 1$$

$$dS_t = \mu S_t dt + \sigma S_t dW_t + S_t \sum_{i=1}^{dN_t} (Y_i - 1), S_0 = s_0$$

Risk-neutral valuation

Applying similar derivations as with Black-Scholes with assumptions to assure that the discounted asset price is a martingale, we get:

$$dS_t = \left(r - \frac{\sigma^2}{2} - \lambda(E^Q[e^Y] - 1)\right) S_t dt + \sigma S_t dW_t + S_t \sum_{i=1}^{dN_t} (Y_i - 1), S_t = s$$

European Call Options

Finally, solving for $f(t, s) = e^{-r(T-t)} E^Q[F(S_T) | S_t = s]$ [CT04] [Mat04], gives:

Jump-Diffusion Formula

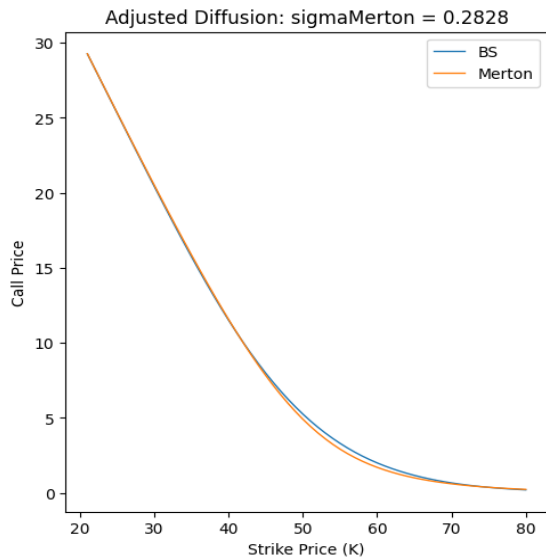
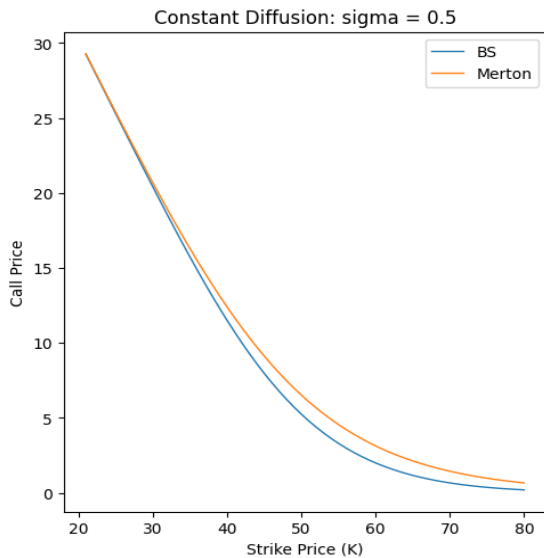
$$f_{Merton}(t, S_t) = \sum_{i=0}^{\infty} \frac{e^{-\bar{\lambda}\Delta t} (\bar{\lambda}\Delta t)^i}{i!} f_{BS}(t, S_t, \sigma = \sigma_i, r = r_i),$$

where $\bar{\lambda} = \lambda e^{\mu + \frac{\delta^2}{2}}$, $\sigma_i = \sqrt{\sigma^2 + \frac{i\delta^2}{\Delta t}}$, and $r_i = r - \lambda(e^{\mu + \frac{\delta^2}{2}}) + \frac{i\mu + \frac{i\delta^2}{2}}{\Delta t}$.

The Kou model also has a closed form formula, $f_{Kou}(t, S_t)$, as well as an approximation via Laplace transformation [Kou07], but we'll focus on the simpler Merton formula.

Note: Put-Call Parity is model-agnostic. Hence, European Put Option pricing can still be derived from Call Option pricing.

Plots



Adjusted Diffusion

- The t-period standard deviations of log-returns for the BS and Merton models [Mat04] can be derived as:

$$SD_{BS}(S_t) = \sigma_{BS} \sqrt{t} \leq \sqrt{(\sigma_{Merton}^2 + \lambda \delta^2 + \lambda \mu^2) t} = SD_{Merton}(S_t)$$

- Hence, Merton prices will always be greater than Black-Scholes prices when $\sigma_{BS} = \sigma_{Merton}$.
 - Intuitively, this is because larger volatility is translated to larger option price.
 - To handle this, we can use the above relationship to set σ_{Merton} w.r.t. σ_{BS} , or vice versa.

Recap #1

1. What are Jump-Diffusion models and how do they work?

- ▶ An attempt to address shortcomings of the BS model via modelling investor sentiment/external market information as jumps.
- ▶ Extensions of the BS model using compound Poisson processes, where many of the same derivation techniques can be applied.

Recap #2

2. How do they compare to the Black-Scholes model?

Black-Scholes

Pros:

- ▶ Simple
- ▶ Versatile

Cons:

- ▶ Many asset features unaddressed

Jump-Diffusion

Pros:

- ▶ Addresses 2 asset features
- ▶ Still somewhat simple
- ▶ Models external information

Cons:

- ▶ Added complexity
- ▶ Additional parameters require tuning
- ▶ Many asset features still unaddressed

Today

- ▶ **Today**, there is a large and growing body of research into Jump-Diffusion models, their extensions, and applications.
- ▶ Certain markets/settings are naturally modelled with sporadic jumps.
 - e.g. Credit risk modelling (Credit events), Insurance claims/losses, etc.
- ▶ Extensions, combinations, and other variations of jump process models have and are continued to be explored to better address other features unaddressed by the Black-Scholes model.
 - e.g. Volatility clustering

References

- [CT04] R. Cont and P. Tankov. *Financial modelling with jump processes*. Financial Mathematic Series. London, UK: Chapman Hall/CRC, 2004.
- [Kou07] S. Kou. *Chapter 2 Jump-Diffusion Models for Asset Pricing in Financial Engineering*. Vol. 15. Handbooks in Operations Research and Management Science. Elsevier, 2007, pp. 73–116.
- [Mat04] K. Matsuda. “Introduction to Merton Jump Diffusion Model”. In: (2004).